

Strategic Games

Review

ERON 403/503

Fall 2022



A strategic game $G = (N, \{A_i\}_{i \in N}, \{u_i\}_{i \in N})$

- N set (finite) of players. $N = \{1, \dots, n\}$
- For each $i \in N$, non-empty set A_i (actions) $A = \prod_{i=1}^n A_i$
- For each $i \in N$, a function $u_i: A \rightarrow \mathbb{R}$,

If for each $i \in N$, A_i is finite, G is a finite

A Nash eq. (NE) of G is a profile $a^* = (a_1^*, a_2^*, \dots, a_n^*)$ such that

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a_i', a_{-i}^*) \quad \forall a_i' \in A_i, \quad \forall i \in N$$

For any $a_{-i} \in A_{-i}$, define $B_i(a_{-i})$ to be the set of i 's best actions against a_{-i} :

$$\begin{aligned} B_i(a_{-i}) &= \{a_i \in A_i : u_i(a_i, a_{-i}) \geq u_i(a_i', a_{-i}) \quad \forall a_i' \in A_i\} \\ &= \operatorname{argmax}_{a_i' \in A_i} u_i(a_i', a_{-i}) \end{aligned}$$

A NE is a profile a^* for which $a_i^* \in B_i(a_{-i}^*)$ for all $i \in N$

Example 5

BoS (Bach or Stravinsky)

	B	S
B	<u>2</u> , <u>1</u>	0, 0
S	0, 0	<u>1</u> , <u>2</u>

$$NE = \{ (B, B), (S, S) \}$$

Prisoners Dilemma

	D	C
D	-1, -1	-4, 0
C	0, -4	-3, -3

→

	D	C
D	3, 3	0, <u>4</u>
C	<u>4</u> , 0	<u>1</u> , <u>1</u>

A strategy s_i is a dominant strategy for player i if

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}) \quad \text{for all } s'_i \in A_i \\ \text{for all } s_{-i} \in A_{-i}$$

	H	T	
H	<u>1</u> , -1	-1, <u>1</u>	No pure strategy equilibrium.
T	-1, <u>1</u>	<u>1</u> , -1	

Consider $G = (N, \{A_i\}_{i \in N}, \{u_i\}_{i \in N})$. Let $\Delta(A_i)$ denote the set of prob. distributions over A_i . $\sigma_i \in \Delta(A_i)$, σ_i is a mixed strategy for player i .

$\sigma_i(a_i) = \Pr(a_i)$, denotes the prob. that σ_i attaches to a_i .

($\sigma_i : A_i \rightarrow [0, 1]$ satisfying prob. axioms)

A profile $\sigma = (\sigma_1, \dots, \sigma_n)$ generates an "action distribution" over A

We assume players choose σ_i 's independently $\Rightarrow \Pr(a | \sigma) = \sigma_1(a_1) \times \dots \times \sigma_n(a_n)$

Player's evaluate σ : $u_i(\sigma) = \sum_{a \in A} \left(\prod_{j=1}^n \sigma_j(a_j) \right) u_i(a)$

$\text{supp}(\sigma_i) = \{a_i \in A_i : \sigma_i(a_i) > 0\}$

$\text{Prob}((a_1, \dots, a_n) | \sigma) = \sigma_1(a_1) \times \sigma_2(a_2) \times \dots \times \sigma_n(a_n)$

Lemma: $G = (N, \{A_i\}_{i \in N}, \{u_i\}_{i \in N})$. If $\alpha^* \in \prod_{i \in N} \Delta(A_i)$ is a mixed strategy NE of G if and only if for every player $i \in N$, every pure strategy (action) in the support of α_i^* is best response to α_{-i}^*

		B	S
α	B	2, 1	0, 0
	S	0, 0	1, 2
		β	$1-\beta$

If Row plays B : 2β

Row plays S : $1-\beta$

If $(\alpha, 1-\alpha)$ is mixed strategy with full support

$$\Rightarrow 2\beta = 1-\beta \Rightarrow \beta = 1/3$$

If $(\beta, 1-\beta)$ has full support

$$\Rightarrow \alpha = 2(1-\alpha) \Rightarrow \alpha = 2/3$$

Mixed Strategy NE = $\left\{ \left(\underset{\text{Row}}{(2/3, 1/3)}, \underset{\text{Column}}{(1/3, 2/3)} \right) \right\}$

Column B : α
Column S : $2(1-\alpha)$

		B	S
α B		2, 1	0, 0
$1-\alpha$ S		0, 0	1, 2

β $1-\beta$

If Row plays B: 2β

Row plays S: $1-\beta$

If $(\alpha, 1-\alpha)$ is mixed strategy with full support

$$\Rightarrow 2\beta = 1-\beta \Rightarrow \beta = 1/3$$

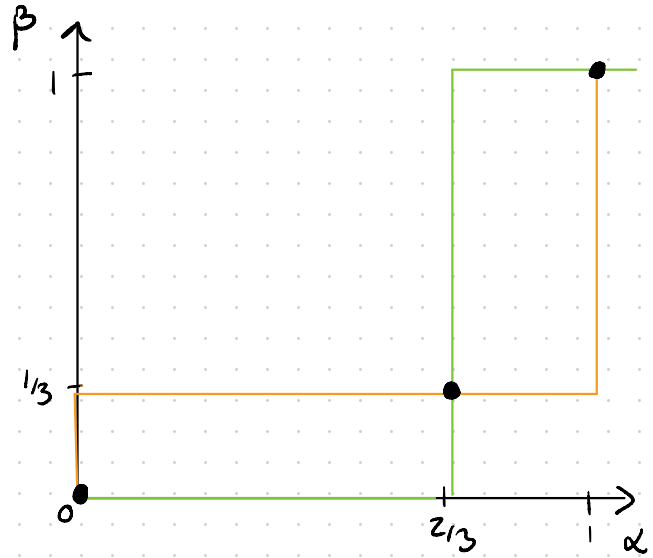
If $(\beta, 1-\beta)$ has full support

$$\Rightarrow \alpha = 2(1-\alpha) \Rightarrow \alpha = 2/3$$

Mixed Strategy NE = $\{(\underset{\text{Row}}{2/3}, \underset{\text{Column}}{1/3}), (1/3, 2/3)\}$

$$\underline{B_R(\beta)} = \begin{cases} \alpha = 1 & \text{if } \beta > 1/3 \\ [0, 1] & \text{if } \beta = 1/3 \\ \alpha = 0 & \text{if } \beta < 1/3 \end{cases}$$

$$\underline{B_C(\alpha)} = \begin{cases} \beta = 1 & \text{if } \alpha > 2/3 \\ [0, 1] & \text{if } \alpha = 2/3 \\ \beta = 0 & \text{if } \alpha < 2/3 \end{cases}$$



$$NE = \{ \overset{1}{\underset{\alpha}{1}}, \overset{1}{\underset{\beta}{1}} \}, \{ \overset{2}{\underset{\alpha}{0}}, \overset{2}{\underset{\beta}{0}} \}, \{ \overset{2}{\underset{\alpha}{2/3}}, \overset{2}{\underset{\beta}{1/3}} \}$$

$$\begin{matrix} \swarrow & \downarrow & \searrow \\ ((1,0), (1,0)) & ((0,1), (0,1)) & ((2/3, 1/3), (1/3, 2/3)) \\ \alpha=1, \beta=1-\beta & & \end{matrix}$$

	L	R
α T	3, 0	0, 1
$1-\alpha$ M	0, 0	3, 1
B	1, 1	1, 0

$\sigma_c(L)$ $\sigma_c(R)$

Suppose Row plays the mixed strategy:

$\sigma_R(T) = \alpha$, $\sigma_R(M) = 1-\alpha$, $\sigma_R(B) = 0$, that is

$$\sigma_R = (\alpha, 1-\alpha, 0)$$

Suppose Column plays any mixed strategy σ_C

Then

$$U_R(\sigma_R, \sigma_C) = 3\alpha \sigma_C(L) + 0 \cdot \alpha \sigma_C(R) + 0 \cdot (1-\alpha) \sigma_C(L) + 3 \cdot (1-\alpha) \sigma_C(R)$$

$$= 3\alpha + 3(1-\alpha) = 3$$

vs

$$U_R(B, \sigma_C) = 1 \cdot \sigma_C(L) + 1 \cdot \sigma_C(R) = 1$$

Since $U_R(\sigma_R, \sigma_C) > U_R(B, \sigma_C) \quad \forall \sigma_C \in \Delta(\{L, R\})$

B is dominated by σ_R

→ We can remove B from Row's consideration.

Knowing Row will not play B, Column now sees that L is dominated by R → Knowing Column will not play L, Row sees that T is dominated by M ⇒ Conclusion the only profile remaining (M, R) is a NE.

This procedure is called: iterated deletion of dominated strategies.

Dominated strategy: We say that for player i , $s_i \in A_i$ is strictly dominated by strategy $\alpha \in \Delta(A_i)$ if for any $s_{-i} \in A_{-i}$ the following holds

$$u_i(\alpha, s_{-i}) > u_i(s_i, s_{-i})$$

Weakly dominated strategy: We say that for player i , $s_i \in A_i$ is weakly dominated by strategy $\alpha \in \Delta(A_i)$ if for any $s_{-i} \in A_{-i}$ the following holds

$$u_i(\alpha, s_{-i}) \geq u_i(s_i, s_{-i})$$

and with at least one strict inequality for one s_{-i}

Nash Theorem: Any finite game has a NE.